

ADVANCED COST ACCOUNTING II ACC322

MARGINAL COSTING

Marginal Costing Versus Absorption Costing

In taking short term, medium term or long term decisions, management of organizations in both the public and private sectors of the economy must adopt some tools that would aid better decision making for the achievement of organizational goals. In Accounting, there are many tools to be used in guiding effective decision making by management. Two of the most popular decision making techniques are marginal costing and absorption costing.

'Marginal Cost', is derived from the word 'margin', which is a well-known concept in economic theories. Thus, quite in tune with the economic connotation of the term, it is described in simple words as the cost which arises from the production of additional increments of output and it does not arise if the additional increments are not produced. This shows that marginal cost is the cost of additional unit of product or service produced in any production process.

From this point of view, marginal costs will be synonymous with variable costs, i.e. prime costs and variable overheads in the short run, but in a way, would also include fixed costs in planning production activities over a long period of time involving an increase in the productive capacity of the business. Thus, marginal costs are related to change in output under a particular circumstance of a case. Marginal costing therefore is about costing an additional product or service on its merit without relating it to the general cost being incurred by the producer in the course of the production process.

The concept of marginal costing is a formal recognition of ideas underlying flexible budgets, break-even analysis and/or Cost-Volume-Profit relationships. It is an application of these relationships which involves a change in the conventional treatment of fixed overheads in relation to income determination.

The concept of marginal costing is based on the important distinction between

product costs and period costs, the former being related to the volume of output and the latter to the period of time rather than the volume of production. Marginal costing regards as products costs, only those manufacturing costs which have a tendency to vary directly with the volume of output. This is in complete contrast to the conventional system of costing under which all manufacturing costs - fixed as well as variable are treated as product costs.

Absorption costing, on the other hand, is the technique of costing that is used in preparing income statements to ascertain the result of operations of private and public sector organizations. The term absorption is about absorbing the general overhead (fixed) costs being shared to all the units of goods or services produced to ascertain total cost of production per unit rather than total variable (marginal) cost per unit.

Absorption costing, therefore, is about costing a product or service with due regards to all the cost elements involved in the production process and, so, appropriate apportionment is to be made for indirect costs to be incorporated into the total cost for each unit of goods or services produced.

Major Differences between Marginal Costing and Absorption Costing

Marginal costing, as explained above, is a technique of costing that considers fixed costs as a period cost and, so, irrelevant when taking decision on the total cost of a product to be compared against its benefit for the determination of its profitability/ viability. It is conceptually about variable or direct costing. Absorption costing, on the other hand, is about appreciating all cost elements as essential for production and sales and, so, all of them are to be captured when determining total cost at all levels of production up to sales.

As the two techniques are used in preparing income statements, some important differences could be noticed when the income statements are compared. The most fundamental difference in the treatment of fixed overhead production cost. While absorption costing accepts the overhead cost as part of the cost of goods sold, marginal costing treats it as a period cost which must be incurred whether or not there is production and, so, it is not to be part of the cost to be aggregated for

a product or service.

The second difference is in the determination of cost of closing stock, where absorption costing recognizes total cost per unit to be multiplied by the closing units to get the value of closing stock while marginal costing recognizes only the total variable cost per unit for that purpose as all fixed costs (production, administrative and selling) must have been covered by contribution margin.

The third difference is in the initial profit shown under each of the two techniques. While absorption costing income statement first shows gross profit before net profit and so on, marginal costing income statement first shows contribution margin, which is the difference between sales and total variable costs. This shows that unless where all the units produced are sold, the two income statements would not amount to the same level of profit at the initial stage.

Absorption costing is used in preparing income statements at the end of an accounting period by all the three forms of businesses (sole proprietorship, partnership and company) in an economy and various governmental establishments. The method is, therefore, better used to aid decision on performance evaluation in respect of result of operations at the end of an accounting period. Marginal costing technique is only relevant for short term tactical managerial decision making, focusing the future happenings of businesses.

Basic Equation, Advantages and Limitations of Marginal Costing

(a) Basic Equation of Marginal Costing

The technique of marginal costing hinges on the contribution made by a product towards fixed costs and profit of the undertaking.

Contribution can be looked at from two angles - either as the total of fixed costs and profit or as the difference between sales and variable costs.

For the sake of convenience, contribution can be stated by way of equation as:
 $\text{Sales} - \text{variable costs} = \text{Contribution}.$

This is sometimes shortened as: $S - V = C$

It is also true that: Contribution - Fixed Cost = Profit

i.e. $C - FC = P$

The formula has been so framed that, if some of the above four factors are known, the remaining one can be easily found out. From this point of view, the equation is of fundamental importance.

Moreover, its significance also lies in the fact that a formula for calculating sales at break-even point has been derived from the basic equation and the formula is:

Sales at break-even point = $\frac{\text{Fixed Cost} \times \text{Sales}}{\text{Sales} - \text{Variable costs}}$

Sales - Variable costs

(b) Advantages of Marginal Costing

- i. **Constant in nature:** Marginal costs remain the same per unit of output irrespective of the volume of production;
- ii. **Facilitating cost control:** The clear-cut division of costs into their fixed and variable components paves the way for a better cost control through flexible budgeting which is based on this important distinction;
- iii. **Simplicity of overhead treatment:** Marginal costing does away with the need for allocation, apportionment and absorption of fixed overheads thereby removing an important source of accounting complications by way of under-absorbed or over-absorbed overheads;
- iv. **Basis for pricing and tendering:** Marginal costing furnishes a better and more logical basis for the fixation of sales prices as well as in tendering for contracts when business is at a low ebb;
- v. **Aid to Profit planning:** The technique of marginal costing enables data to be presented to management in a manner as to show cost-volume profit relationships; and
- vi. **Realistic valuation:** Elimination of fixed overheads from the cost of production means that finished goods and work-in-progress are valued at their marginal cost and, therefore, the valuation is more realistic and uniform as compared to the one when they are valued at their total cost.

(c) Limitations of Marginal Costing

- i. ***Difficulty in analysis:*** Considerable difficulty is always experienced in analyzing overheads into their fixed and variable components;
- ii. ***Lop-sided Emphasis:*** Marginal costing has a tendency to attach more importance to the selling function which has the effect of relegating the production function to a comparatively unimportant position. However, the efficiency of a business is to be judged by taking together its selling as well as production functions into account;
- iii. ***Difficulty in application:*** The technique of marginal costing cannot be adequately applied in the case of industries in which, according to the nature of business, large stocks have to be kept in the form of work-in-progress;
- iv. ***Limited Scope:*** As marginal costing distinguishes between the treatment of fixed and variable components of costs, it is difficult to adopt the technique in capital-intensive industries where fixed costs are very large; and
- v. ***Inappropriate basis for pricing:*** Selling price cannot reasonably be fixed on the basis of contribution alone.

PREPARATION OF INCOME STATEMENTS

Income Determination: Under Absorption and Marginal Costing

According to traditional costing system, fixed costs of production are assigned to products to be subsequently released by way of expenses as part of cost of goods sold or are carried forward as part of the cost of inventory. Such an approach to the treatment of fixed costs has brought into attention various methods of allocation of overheads to different departments on an equitable basis and their proper apportionment to units produced.

Illustration 1 (Comparative Income Statements)

Green-Green Limited furnishes the following details for the year ended 31 December 2021 for preparing the income statement for the year:

Sales	1,000 units @ N10 per unit
Manufacturing costs	N2,200
Variable manufacturing cost	1,100 units @ N6 per unit
Inventory at close	100 units
Fixed selling & admin. Expenses	N500
Variable selling & admin. expenses	N400

Role of contribution

Contribution is of vital importance for the system of marginal costing. The rationale of contribution lies in the fact that, where a business manufactures more than one product, the profit realized on individual products cannot be possibly calculated due to the problem of apportionment of fixed costs to different products which is done away with under marginal costing. Therefore, some methods are required for the treatment of fixed costs and marginal costing and answer to the challenge is 'contribution'.

Contribution is the difference between sales and the variable cost of sales and is, therefore, sometimes referred to as a 'gross margin'. It is visualized as some sort of a 'fund' or 'pool' out of which all fixed costs, irrespective of their nature, are to be met and to which each product has to contribute its share. The difference between contribution and fixed costs is either profit or loss as the case may be.

The concept of contribution is useful in the fixation of selling prices, determination of break-even point, selection of product mix for profit maximization, make or buy decisions, acceptance or rejection of special orders and ascertainment of the profitability of products, departments, etc.

Illustration 2. Contribution Margin

The following information has been gathered from the accounts of Prudent Engineering Limited, organized on departmental basis:

Departments

	<u>X</u>	<u>Y</u>	<u>Z</u>
	N	N	N
Variable costs	90,000	100,000	150,000
Fixed costs:			
(apportioned on the basis of sales)	<u>20,000</u>	<u>30,000</u>	<u>50,000</u>
Total Costs	110,000	130,000	200,000
Sales	<u>100,000</u>	<u>150,000</u>	<u>250,000</u>
Profit (Loss)	<u>(10,000)</u>	20,000	50,000

With the object of doing away with the 'X' department incurring a loss, the management asks for your opinion on:

- a. The closure of Department X on the basis of the above information.
- b. The comparative profitability of different departments if specific fixed costs are ascertained to be ₦5,000 for Department X, ₦55,000 for Department Y and ₦30,000 for Department Z, the remaining ₦10,000 being general fixed costs.

Required:

Prepare appropriate statements so as to help management in arriving at a decision on the above points. Also give your comments, explaining the position presented in the statements.

Relevant Cost and Relevant Revenue

Environment affected by limiting factors often requires consideration of relevant costs and relevant revenues in taking short term decisions. For instance, decisions involving application of marginal costing such as special order pricing, make or buy decision etc often make use of relevant costing. The CIMA Official Terminology defines Relevant Costs/Relevant Revenues as "Costs and revenues appropriate to a specific management decision.

The three major characteristics of relevant costs are:

- They are future cost/future revenues
- They are differential costs/revenues that differs between alternative decisions
- This involve cash flows

Relevant costs of elements of cost

1. **Relevant Cost of Material:** Where materials in stores will be used to implement decision, relevant cost is NIL, value. However, if materials in stores have replacement or market value or can be put into an alternative use, then, the values involved are relevant.
2. **Relevant Cost of Labor:** Fixed salaries of workers are irrelevant, only the associate opportunity cost of fixed salary earner is relevant.

ILLUSTRATION 1

A firm received a special order. The order will utilize material worth of ₦10,000 in stores. The material has no market value and cannot be put into an alternative use. The order will require additional material CD to be procured for ₦30,000. Factory workers will be paid a total of ₦50,000.

The order will also make use of an idle plant that has a written down value of ₦200,000. Depreciation is calculated at ₦5,000 per month. The order will take one month to complete.

Required: Determine the minimum price to charge.

Computation of Minimum Price

₦	
Existing Stores	Material in Nil Value
Material CD	30,000
Factory workers' wages	50,000
Manager's salary	Nil Value
Plant depreciation	Nil Value
Minimum Price	<u>80,000</u>

Bases for Computation:

- Existing materials in stores are sunk cost
- Manager's salary considered as irrelevant/sunk
- Plant depreciation on an idle plant – sunk cost
- Relevant cost includes material CD value and factory workers' wages.

Activity Based Costing (ABC)

Activity based costing is a method of cost attribution to cost units on the basis of benefits received from indirect activities. The performance of particular activities and demands made by the activities on the resources of organization are linked together so that the cost of product is arrived at as per the quantum of activities performed to produce a product or render a service.

Activity based costing may be defined as a technique which involves identification of costs with each cost driving activity and making it as the basis for apportionment of costs over different products or jobs. That is, a costing method that identifies activities in an organization and assigns the cost of each activity to all product or services according to the actual consumption by each. The Chartered Institute of Management Accountants defines it as a "technique of cost attribution to cost units on the basis of benefits received from indirect activities e.g. ordering, setting up, and assuring quality

Significant Terms

1. **Cost Objects.** Generally, the products are cost objects, but the customers, services or locations can also be the cost objects.
2. **Activities.** These consist of the aggregate of different tasks and are concerned with functions associated with cost objects. There are two types

of activities-

(A) Support activities, (B) Production process activities.

Support activities are, schedule production, set up machine, purchase materials, inspect items, customer orders, supplier records etc. Under the production process activities machine products and assembled products are included.

Cost Pool. It is another name given to a cost center and, therefore, an activity cost center may also be termed as an activity cost pool.

Cost Drivers. The causes for incurrence of overhead cost are known as cost drivers. A cost driver is a factor the change of which results in a consequential change in the total cost of a related object.

Examples of cost drivers :

1. Machine Setups 7. Maintenance requests
2. Purchase orders 8. Machine time
3. Quality Inspections 9. Power consumed
4. Production orders (Scheduling) 10. Kilometers driven
5. Material Receipts 11. Beds occupied
6. Inventory movements 12. Flight-hours logged

The activity cost drivers can broadly be classified into following three categories :

- A. Transaction drivers ;
 - B. Duration drivers ;
 - C. Intensity drivers ;
- A. **Transaction drivers.** For example, the purchase orders processed, customer orders processed, inspections performed and the set-ups undertaken, all count the number of times an activity is performed.
 - B. **Duration drivers.** Mean the amount of time required to perform an activity. Examples of duration drivers are set-up hours and inspection hours.
 - C. **Intensity drivers.** Refer to drivers which directly charge for the resources used each time as activity is performed.

Activity Based Costing Versus Traditional Costing

Following are the main differences between activity based costing system and traditional costing system :

1. Under ABC system, overhead costs are identified to each major activity instead of the department as under traditional costing system.
2. The term 'cost driver' is not used under traditional costing system. Popular terms are basis of allocation or apportionment. Under ABC system, cost drivers are fewer in number for the purpose of charging overheads to products.
3. ABC system uses separate rates for support centers and there is no reallocation to production centers, as is the case under traditional costing system.

Thus, traditional costing system reveals less accurate costs as compared to ABC system. This is primarily because cause and effect relationship is considered under ABC system to identify support cost objects, which is not the case under traditional costing system.

ABC system requires the following :

1. Setting-up of an information system which could help trace all the costs to cost objects.
2. Support from top to bottom because the system involves people at all levels.
3. Integration of system into financial system. For it, computerization may be required.

Illustration. 1 : A company manufacturing two products furnishes the following data for a year

Product	Annual Output	Total Machine hours	Total number of purchase orders	Total number of set-ups
A	5,000	20,000	160	20
B	60,000	120000	384	44

The annual overheads are as under : #

Volume related activity costs	5,50,000
Set up related costs	8,20,000
Purchase related costs	6,18,000

You are required to calculate the cost per unit of each Product A and B based on :

- (i) Traditional method of charging overheads.
- (ii) Activity based costing method.

COST-VOLUME-PROFIT ANALYSIS

The Concept of Cost-Volume-Profit

Cost-Volume-Profit (CVP) analysis involves the analysis of how total costs, total revenues and total profits are related to sales volume, and is, therefore, concerned with predicting the effects of changes in costs and sales volume on profit. It is an analysis that guides decision making in respect of going into a production process, expanding an existing production level or diversifying into unusual areas of operations.

CVP analysis seeks to find out answers to the following questions:

- a. What would be the cost of production under different circumstances?
- b. What has to be the volume of production?
- c. How many units of each product can be manufactured in order to earn target profit?
- d. What is the difference between the selling price and cost of production?
- e. What are the effects of a percentage increase or decrease in cost on production volume and on profit;

The concept of Break-even Analysis

The C-V-P analyses short-term profit planning that establishes relationship among costs, volume and profit. Even though profit determination is important, it is equally necessary to find out production capacity utilization that enables a firm to break even in order to enable the firm determine the viability of its business. The C-V-P Analysis is broad, while the narrow concept of C-V-P analysis is referred to as Break-even point analysis.

The Break-Even-Point is the level of production activity at which there is neither profit nor loss. At Break-even-point, BEP, total cost incurred for a product equates total revenue realized for selling the product.

Thus at BEP:

Total Cost = Total Revenue

i.e $TC = TR$ or $TR - TC = \text{Zero Profit}$

where $TC = \text{Total Fixed Cost} + \text{Total Variable Cost}$.

After the BEP, fixed cost is no longer incurred. The cost incurred after BEP is

limited to variable cost.

Assumptions of Break-even Analysis

Break-even-analysis and Cost-Volume-Profit analysis are based upon certain assumed conditions which are to be rarely found in practice. Some of these basic assumptions are as follows:

- a. The principle of cost variability is valid.
- b. Costs can be resolved into their fixed and variable components.
- c. Total Fixed costs remain constant throughout the relevant range.
- d. Variable costs per unit vary proportionally with volume.
- e. Selling price does not change as volume changes.
- f. There is only one product or, in the case of multiple products, sale mix remains constant.
- g. There will be no change in general price level.
- h. Productivity per worker remains mostly unchanged.
- i. There is synchronization between production and sales, i.e. no inventory
- j. Revenue and costs are being compared with a common activity base e.g. sales value of production or units produced.
- k. The efficiency of plant can be predicted.

A change in any one of the above factors will alter the break-even point so that profits are affected by changes in factors other than volume. Thus, the break-even chart must be interpreted in the light of the limitations of underlying assumptions, especially with respect to price and sale mix factors.

Computation of Break-Even-Point, BEP

BEP can be computed through different mathematical approaches. However, the two most popular mathematical approaches are: (i) equation technique or (ii) contribution margin technique.

- (i) *The equation technique:* This technique uses the following formula which also expresses the relationship of the items of income statement.

Sales at BEP = Variable Expenses + Fixed Expenses + Zero Profit
or TR = TC + Zero Profit.

This simple equation may be adapted to any break-even or profit estimate situation.

- (ii) *Contribution margin or marginal income technique:* This is obviously based on the concept of marginal costing. Contribution margin is the difference between sales and variable expenses. Where break-even point is desired, sales and expenses are analyzed as thus:

Unit selling price - Unit variable expenses = Unit contribution and,
Contribution – Fixed Cost = Profit

This unit contribution margin is divided by total fixed expenses to secure the number of units which have to be sold to break-even.

$$\text{i.e. } \frac{\text{Fixed Expenses}}{\text{Contribution per unit}}$$

These two techniques can be illustrated by an example.

Illustration 1: Mathematical Approach

Alhaji plans to sell toy rockets at Kano International Trade Fair. He may purchase these rockets at ₦5 each, with the privilege of returning all unsold rockets. The booth rent at the fair is ₦2,000 payable in advance. The rockets will be sold at ₦9 each.

Determine the number of rockets, which must be sold to break-even as well as the number of rockets to be sold to yield a 20 per cent operating margin on sales.

*Determine the production volume or production units that will realize 20% profit margin?

Other simple formulae that can be used to determine cost-volume-profit relationship are as follows:

$$\text{i. Break-even point (in ₦)} \\ = \frac{\text{Fixed Cost}}{\text{Contribution Margin Ratio, (CMR)}}$$

OR C/S Ratio

Note that C/S Ratio is also termed as Profit-Volume Ratio i.e P/V Ratio Where:

$$\text{C/S ratio} = \frac{\text{Contribution per unit} \times 100}{\text{Sales price per unit}}$$

- ii. Level of sales to result in target profit (in units)

$$= \frac{\text{Fixed Costs} + \text{Target Profit}}{\text{Contribution per unit}}$$

Profit-Volume Ratio, P/V Ratio

Profit Volume Ratio, P/V Ratio or Contribution Margin Ratio, CMR establishes the relationship between contribution and sale revenue. It determines the profitability of a product. It can be calculated in different ways below:

$$1. \text{ P/V Ratio} = \frac{\text{Unit Contribution}}{\text{Unit Sales Price}} \quad \text{i.e.} \quad \frac{S - V}{S}$$

$$2. \text{ P/V Ratio} = \frac{\text{Total Contribution}}{\text{Total Sales Revenue}} = \frac{\text{TSR} - \text{TVC}}{\text{TSR}}$$

Where: S - indicates unit contribution

V - Indicates unit variable cost

TSR - indicates total sales revenue

TVC - indicates total variable cost

$$3. \text{ P/V Ratio} = \frac{\text{FC} + \text{Profit}}{\text{Total Sales}}$$

Illustration 2.

Poly Nigeria Limited manufactures and sells a unique product, the selling price of which is ₦20. The summarized income statement for last year is as follows:

Sales	₦	₦
		800,000
Direct materials		120,000
Direct wages		160,000
Variable production overhead	80,000	
Fixed production overhead	100,000	
Fixed Administration overhead	75,000	
Fixed Selling and Distr. Overhead	<u>60,000</u>	<u>595,000</u>
Net Profit before tax		205,000
Less: provision for taxation (40%)		<u>82,000</u>
Net Profit after tax		<u>123,000</u>

Required:

- a. Calculate the break-even point for last year.
- b. What do you understand by the terms 'profit volume ratio' and 'margin of safety'. Illustrate using last year's result.
 - a. Determine the number of units to sell in the current year to achieve an after-tax profit of ₦150,000
 - b. Calculate the sales value required to achieve a net profit before tax of 15% of total revenue.

C.V.P Graphs

In addition to determining break-even value and P/V Value mathematically, some business organizations usually at the same time present C-V-P graphs in order to enable managers appraise the business profitability at a glance.

Generally, C-V-P graphs can be split into different component graphs. However, this study only covers two types of C-V-P graphs that include:

- i. Break-Even Point Graph, and
- ii. Profit-Volume, P/V Graph

The Break-Even Point Graph

A break-even chart shows the level at which total cost incurred for a product equates total revenue realized. The break-even point is obtained where total cost graph intersects with the total revenue graph. The costs and revenues are indicated in the Y-axis while output in units is indicated in the X-axis.

The total cost line is obtained from: $\text{Total Fixed Cost} + \text{Variable Cost}$ at different production units

While:

The total revenue line is obtained from: Total revenue at each level of output, which is the product of output multiplied by unit selling price.

Illustration 3: Graphical demonstration

A company makes and sells a single product. The variable cost of production is ₦3 per unit, and the variable cost of selling is ₦1 per unit. Fixed costs total ₦6,000 and the unit selling price is ₦6. The company budgets to make and sell 3,600 units in the next years.

Required: A break-even chart and later to plot (i) Profit-Volume graph, with

each graph showing the expected amount of output and sales required to break even, and the margin of safety in the budget.

Solution 3 (i) Break-Even Chart

The break-even chart shows:

- Fixed cost line of ₦6,000,
- Total costs line, with variable cost added to fixed cost line, and
- Total revenue

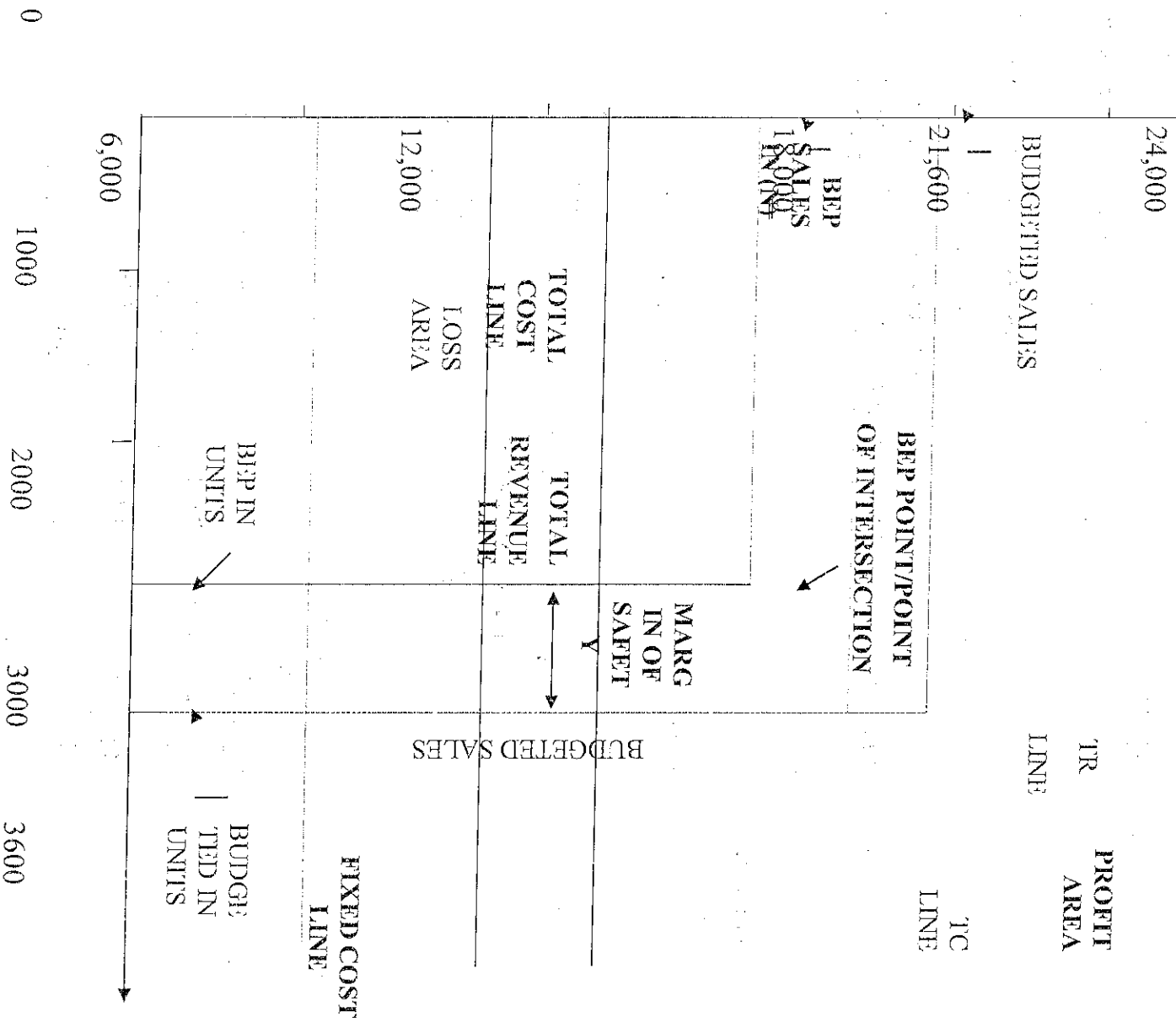


Figure 1: Break Even Chart

OUTPUT/SALES IN UNITS

Note: At BEP, TC Line = TR Line = Point of intersection
 = 3,000 units sold for ₦18,000 At BEP, Profit = Zero

i.	M/S from the graph:	Revenue in:	
		₦	Units
	Budgeted	21,600	3,600
	Break-Even Point	18,000	3,000
	Margin of Safety	3,600	600

Thus, the sales revenue can be fall by ₦3,600 sales revenue or by 600 units and the firm can still break even with zero profit.

BUDGETING AND BUDGETARY CONTROL

At the beginning of the financial period of every organization, whether the organization is At the beginning of the financial period of every organization, whether the organization is

Definition of Budget

A budget is a quantitative and financial plan of the operations of an organization or an individual. It identifies the resources and the commitments required to fulfill the organization's or individual's goals for the budgeted period. Thus, a budget includes both financial and non-financial aspects of the planned operations. The process of preparing budgeting is termed budgeting.

A budget has the following features:

- a) It is a plan in quantities and financials
- b) It must have objectives, and a clearly expressed means of achieving the stated objectives
- c) The means to achieve the objectives are the resources required to fulfil those objectives. Thus there must be definite resources set aside to achieve the set objectives.
- d) The set objectives must be achieved within a particular time frame. This time frame is termed the budget period.

Types of Budgets

There are a variety of budgets; some of them are as stated below:

Long Term Budgeting

This is also referred to as long range planning, strategic or corporate planning. It is a systematic and formalized process for purposely directing and controlling future operations towards desired objectives for periods extending beyond one year.

The planning process involves the following sequential steps :

- a) Identify objectives
- b) Search for alternative courses of action
- c) Gather data about the various alternatives
- d) Evaluate the alternatives on the basis of cost-benefit relationships
- e) Select the optimum alternative and formulate these into plans
- f) Implement the long range plans through annual budgets
- g) Monitor the actual results
- h) Respond to variances

Short Term Budgeting

These are operational plans which generally cover one year and which form the basis of the annual budgets. They are detailed, containing well-defined targets, which translate into work plans and performance measurements for each unit in the business. Short term budgets are a means of implementing strategic plans.

These are developed after due consideration of:

- The prevailing business environment viz: internal and external
- The available resources viz: physical, human and financial.

Annual Budgets

The annual budget translates the company's long-range plan into short-term operating targets. These are usually developed by involving all people responsible for achieving the targets and objectives of the company.

Functional Budgets

The functional budgets are the budgets developed for the operations of each

function in the organization. Usually, the budget for the function that has the binding or effective budget factor or limiting factor is first developed bearing in mind the budget factor.

All the other functional budgets are subsequently developed from the budget of the function with the binding constraint. Most often, sales is the limiting budget factor. However, this is not necessarily always the case. The following are examples of functional budgets:

- *Sales budget in quantity and in value. Other sales related budgets
 - i. Selling and distribution cost budget
 - ii. Advertising cost budget
 - iii. Sales promotion cost budget etc.

- *Production budget in quantity and cost, other production related budgets;
 - iv. Labor usage and cost budget
 - v. Materials usage and cost budget
 - vi. Material purchases budget
 - vii. Production overhead budget.
 - viii. Production cost budget

- *Service function budgets
 - vii. Administration cost budget
 - viii. Personnel cost budget

- *Policy Related Budgets
 - ix. Capital expenditure budgets
 - x. Research and development budgets
 - xi. Advertising and promotional budget
 - xii. Training and staff development budgets

The Cash Budget: The cash budget summarizes the cash flow effects of all of the functional and policy related budgets by coordinating them into one summarized statement.

Master Budget: The master budget summarizes all the budgets into a coordinated projected financial statement comprising of:

- budgeted trading, profit and loss account (Income Statement) or income and expenditure account as the case may be;

- budgeted balance sheet or statement of affairs; and
- budgeted cash flow statement for the period.

Principal Budget Factor

The principal budget factor or limiting factor or key budget factor is a factor, which at any time is an overriding planning limitation on the activities of the organization. Examples of principal budget factors include: Staffing limitations, scarcity of materials and other logistics, limited financial resources, low sales demand, limited storage facilities, etc.

The Objectives of Budgeting

Budgeting is a critical tool set out to:

- a) Aid the planning of annual operations;
- b) Coordinate the activities of the various parts of the organization and to ensure that the parts are in harmony with each other;
- c) Communicate plans to the various responsibility centre managers;
- d) To motivate employees to strive to achieve target; To control activities; and
- e) To evaluate the performance of managers.
- f) To control activities; and
- g) To evaluate the performance of managers.

Each of these objectives is further explained below:

Planning

Budgets compel or force planning to take place. The result is that problems are anticipated before they arise and hasty decisions made on the spur of the moment based on expediency rather than reasoned judgment will be minimized. The annual budget also leads to the refinement of strategic plans.

Co-ordination

The budget is a vehicle that synchronizes and harmonizes the separate plans of the different units, sections and segments of the organization. Budgeting compels an examination of the relationships between various operations and hence the resolution of any conflict.

Communication

Top management communicates its expectations to lower management through the budget, which is an executive order. Much vital information about the

organization is communicated through the process of preparing the budget. The budgetary reports are a feedback to management on the performance levels of each unit of the organization.

Motivation

Realistic and attainable budgets serve as a standard of good performance and thus motivate staff to strive to achieve them. However, for a budget to motivate, all staff must participate in its development.

Control

Variance reports generated from the budgetary control system are a good control mechanism facilitating management by exception, which is efficient and cost effective. Budgets clarify the authority and responsibility of managers by setting budget limits to their spending powers and setting targets to be achieved.

Performance Evaluation

Performance could be evaluated by measuring the success of achieving budgeted results. The budgets based on the accepted standards are the benchmarks for measuring performance.

The Budget Committee

The Budget Committee is made up of members of senior management that oversees all budgetary matters. A typical budget committee includes the chief executive officer, heads of strategic business units and the chief finance officer. The committee sets or approves the overall budget goals for the organisation, and its major business units, directs and coordinates budget preparation, approves the final budget, monitors operations as the year unfolds, and reviews the operating results at the end of the period.

The budget committee also approves major revisions of the budget during the period. This committee usually consists of sectional or departmental managers and is usually serviced by the Budget Officer who normally is the finance officer. Apart from the above, the functions of the committee may include the following:

- a) Determination of the Budget Policy Guidelines and selection of budget policies compatible with organizational goals and objectives;
- b) Establishment of the budget timetable;
- c) Review of budget estimates submitted by sectional heads;
- d) Facilitating the co-ordination of the budgets;

- c) Suggesting amendments to budgets and revising budget estimates when necessary;
- f) Approval of budgets after amendments;
- g) Facilitating the generation of budgetary control reports;
- h) Analyzing budget reports and recommending changes where necessary;
- i) Examination of variances, suggesting investigation of variances and recommending solutions to remedy off-standard performance; and
- j) Offering advice to top management on all matters concerning the budget.

Illustration .1

Asuo Ltd, plans to produce and sell 8,000 sachets of sugar during the year 2023. The selling price is expected to be #30 per sachet.

1 unit of sachet requires 4 units of Material A, 3 units of Material B and 2 units of Material C. Opening stock of raw materials is as follows:

	Units	Amount
		#
Material A	45,000	270,000
Material B	35,000	70,000
Material C	25,000	50,000

The closing stock for each raw material is to be at a level which would meet the production requirement of 7,000 sachets of sugar. There is no opening or closing stocks of sugar.

Purchase prices for all raw materials during the year 2023 are expected to be 30% higher than the prices reflected in the opening stock values.

Sales and purchases are on credit and the opening balances being as follows:

	#
Debtors	350,000
Creditors	270,000

The Company expects to receive #525,000 from debtors during the period and to pay #235,000 to creditors

Required:

Prepare a budget for raw material purchases and calculate the closing balances for debtors and creditors

Illustration 2

Sakyi Ltd. produces two products, Cake and Bluna, which use the same raw materials, oil and fat. One unit of Cake uses 3 liters of oil and 4 kgs of fat. One unit of Bluna uses 5 liters of oil and 2 kgs of fat. A liter of oil is expected to cost #3,000 and a kg of fat #7,000.

Budgeted Sales for 2021 are 8,000 units of Cake and 6,000 of Bluna; finished goods in stock at 1/1/2021 are 1,500 units of Cake and 300 units of Bluna, and the company plans to hold stocks of 600 units of each product at 31/12/2021.

Stock of raw materials are 6,000 liters of oil and 2,800 kgs of fat at 1/1/2021, and the company plans to hold 5,000 liters and 3,500 kgs respectively at 31/12/2021.

The store's manager has suggested that a provision should be made for damages and deterioration of items held in store as follows:

Cake	-	loss of 50 units
Bluna	-	loss of 100 units
Oil	-	loss of 500 litres
Fat	-	loss of 200 kgs

You are required to:

- (a) Prepare a material purchases budget for the year 2021
- (b) Distinguish between Functional Budgets and Master Budgets

The Meaning of Budgetary Control

Budgetary control is the whole system of control established in organizations to plan the activities of the organization and take steps to achieve the set plans. The control system begins with the determination of standards, the formulation of budgets, comparing actual results with standard performance to determine variances and analyzing the variances. Material variances are then investigated and corrective actions taken.

Objectives of Budgetary Control

Budgetary control is usually setup in order to:

- a) aid the planning of annual operations;

- b) coordinate the activities of the various parts of the organization and to ensure that the parts are in synchrony or harmony with each other;
- c) communicate plans to the various responsibility center managers;
- d) control activities;
- e) motivate employees to achieve targets; and
- f) evaluate the performance of managers.

The Budgetary Control Process Involves the Following Processes:

- g) Establishment of budgets for various budget centers. A budget center is a clearly defined part of an organization for the purpose of operating a budget system;
- h) Creation of Control periods. A budget period is the period for which a budget is prepared and over which such control aspects take place. The following factors influence what control period will be in use;
 - i. The nature and size of the business,
 - ii. The part of the business for which the budget is being prepared,
 - iii. The basis of control, and
 - iv. The velocity of operations, etc.
- i) Keeping records of Actual results through the financial accounting system;
- j) Comparing the actual results with the budget;
- k) Variance Analysis and Reporting on the variances analyzed; and
- l) Investigation and reporting on the causes of variances.

STANDARD COSTING TECHNIQUES

Standard cost is a predetermined calculation of how much cost should be under specified working conditions. In other words, standard cost is a standard expressed in money terms. It is built up from an assessment of the value of cost elements. Its main uses are providing bases for performance measurement, control by exception reporting, valuing stock and establishing selling prices

Types of standards

There are four types of standards; namely: basic standards, ideal standards, attainable standards and current standards.

a) Basic Standards

These are long term standards remaining unchanged over several years. They are standards established for use over a long period from which a current standard can be developed

The uses of basic standards include:

- i. To show trends over time for such items as:
 - materials prices
 - labour rates
 - efficiency levels and the long term effects of changing methods

- ii. Used as a basis for setting current standards. Basic standards cannot be used to highlight current efficiency or inefficiency and would not normally form part of the reporting system except as a background statistical exercise.

b) Ideal Standards

These are standards which can only be attained under the most favourable working conditions. Ideal standards are based on optimal operating conditions such as:

- no break downs of machinery
- no wastage of materials
- no stoppages
- no idle time
- no spoilage
- no shrinkage of materials etc.

The uses of Ideal Standards include:

- i. Used as long term targets
- ii. Used for long term development purposes and investigative purposes

c) Attainable standards

These are standards which can be achieved if a standard unit of work is carried out efficiently, a machine properly operated or material properly used. Allowances are made for normal losses and machine breakdowns. It represents future performance and objectives which are reasonably met

Uses of attainable Standards include:

- i. Frequently used for routine control with the purpose of providing tough but realistic targets

- motivating staff and management to achieve targets
- ii. Used for stock valuation for purposes of:

- product costing
- cost control

Attainable standards should be revised periodically to reflect:

- new conditions
- new prices, methods, technology, etc.
- d) Current Standards

These are standards set for use over a short period to reflect current conditions. Where conditions are stable then current standards will be the same as attainable standards. But where a temporary problem exists then a current standard could be set to deal with the problem. Uses of current standards

- i. where there is a temporal problem with material quality
- ii. unexpected price rise due to inflationary circumstances
- iii. energy crises
- iv. significant standards containing some subjective elements etc..

Objectives of Standard Costing

- a. To provide a basis for assessing performance and efficiency
- b. To control costs by establishing standards and the analysis of variances and ascertaining the reasons for the variances
- c. To facilitate the practice of management by exception at the operational level
- d. To assist in formulating budgets
- e. The standard costs can be used as ends for:
 - valuing stocks and work in progress
 - Profit planning and decision making
 - pricing policy (especially where cost plus pricing systems are used)
- f. to motivate staff and management

Merits of Standard Costing

- a. Assists in assessing performance and efficiency
- b. Enhances cost control by encouraging re-appraisal of methods, materials and

techniques

- c. Makes budgeting easier
- d. Facilitates the prudent practice of management by exceptions
- e. It provides guidance on possible ways of improving performance
- f. Provides a basis for estimation and forecasting
- g. It is a source of motivation (if full participation and involvement is sought)
- h. Simplifies stock valuation and pricing policy
- i. Assigns responsibility for non-standard performance

Limitations of Standard Costing

- j. The philosophy of standard costing is challenged as being inappropriate
- k. Standard costing can only exist realistically within the frame work of a budgetary system
- l. Variance analysis merely directs attention to the cause of off-standard performances. It does not solve the problem nor does it establish the reasons behind the variance. Solutions to the identified problems are management tasks
- m. All standards involve forecasting which is subjective with the inherent possibility of error.
- n. the process is a bit complex and difficult and is not understood by most line managers
- o. variance analysis are post mortems on past events and does not correct that past error or wrong.
- p. it may be expensive and time consuming to install
- q. in volatile conditions with rapidly changing methods, rates and prices, standards quickly become out of date losing their control and motivational effects
- r. the usefulness of a number of variances is questionable

Setting Standard Costs

To be able to set standards, we take every element that contributes to the operations of an organization and set standards for each of them. We consider the following:

- the types, and prices of material and parts
- the grades, rates of pay and time for the labour
- the production methods and layouts

- the tools, jigs and machines to be used

Setting Standards for Materials

- a. The materials content of each product is derived through engineering studies. Examples may include:
 - i. raw materials
 - ii. Sub-assemblies
 - iii. piece parts
 - iv. finishing materials
 - b. Determine the quantities required of each of the inputs, making allowance for normal losses in production arising from machining loss, e.g.
 - i. Evaporation
 - ii. expected breakages
 - iii. expected rejections, etc.
 - c. Obtain material prices from the purchases or procurement department
 - i. these prices should be the forecast or expected prices for the relevant budget period and not past costs
 - ii. the prices should reflect:
 - trend in material prices
 - anticipated changes in purchasing policies
 - quantity and cash discounts
 - carriage and packing charges
- On the basis of the variables obtained above, you then set standards for materials as standard quantities of each material at the standard material price per unit.

Setting Standards for Labour

- a. Specify the exact grades of labour to be used. This can be by the use of work study projections, techniques of work measurement, etc.
- b. Determine the time to be taken for a unit of output. The standard hour which is the quantity of work achievable at standard performance in an hour or minute
- c. the personnel department should make a forecast of the relevant wage rates for the control period
- d. set the standard labour cost as standard hour at standard wage rate

Setting Standards for Overheads

The predetermined overhead absorption rates are the standard rates for overheads for each cost center using the budgeted standard labor hours as the activity base

$$VOR = \frac{\text{Budgeted variable overheads for cost center}}{\text{Budgeted standard labor hours for cost center}}$$

$$FOR = \frac{\text{Budgeted fixed overheads for cost center}}{\text{Budgeted standard labor hours for cost center}}$$

Where: VOR is Variable Overhead Rate, and

FOR is Fixed Overhead Rate

Setting standard for selling price

This activity is a top management activity, which is done after considering the following factors:

- c. Anticipated market demand
- a. competing products
- b. Inflation estimates
- c. elasticity of demand, etc

VARIANCE ANALYSIS

A variance is the difference between a standard level of performance that ought to have been achieved and the actual activity achieved.

Analysis of Variances: Variance analysis is the breaking down of the total profit variance into its smallest parts.

Illustration I

Ghartex Ltd. specializes in the manufacture of textiles for export and the local market. The following information relates to the Company's budget for a given month.

Budgeted Production Qty:

120000 meters direct material

Material A 0.5 kg (#1,200 per kg)

0.5 kg (#1,600 per kg)

30 mins. (#6,000 per hour)

#30 million

- #15 million

Production overhead is absorbed on the basis of direct Labor hours. Actual Results achieved during June 2000 are as follows:

Production	130,000 meters	Director Material Purchased.
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80,000 kg	Costing	#88 million
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Material B	70,000 kg	Costing	#98 million
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Direct Labour Hours 80,000 hours

worked costing #400 million. Variables

overheads - #28 million

Fixed Overheads - #18 million

Required: Compute the various variances from the above data.

The basic cost variances are the difference between standard cost allowed for actual output and the actual cost incurred for that actual output. The total cost variance can further be analyzed into variances for each of the elements of cost, namely: material cost variance, labor cost variance, variable overheads variance and fixed overheads variance.

1. Material Cost Variances, MCV

This is the difference between standard materials cost and actual materials cost of the actual production volume achieved. The material cost variance can further be analyzed into materials price variance and materials usage variance. The formula is:

$$\text{MCV} = (\text{SQ of production achieved} \times \text{SR}) - (\text{Actual quantity used} \times \text{Actual unit price})$$

Note: Actual quantity used = Actual quantity purchased where there is no opening and no closing inventories.

Where:

SP denotes unit selling price

SQ denotes quantity use in producing one

SQ of production achieved = $SQ \times \text{Actual output}$

a. **Direct Material Price Variance**

This is the difference between the standard price and actual purchase price of the actual quantity of materials purchased or used. The formula is:

$$AQP (SP - AP)$$

Where:

$AQP = \text{Actual quantity}$

$purchased AP = \text{Actual unit price}$

b. **Direct Material Usage Variance**

This is the difference between the standard quantity specified for the actual production and the actual quantity used for the actual production expressed at standard price. The formula is:

$$SP (SQ \text{ of production achieved} - AQ \text{ used})$$

Where AQ used denotes Actual quantity used

c. **Labour Cost Variances**

This is the difference between the standard direct labour cost and actual direct labour cost for the actual production achieved. This variance can further be analysed into labour rate variance and labour efficiency variance.

d. **Labour Rate Variance**

This is the difference between the standard direct labour rate and actual direct labour rate per hour for the actual total hours worked and paid for.

Possible Causes of direct labour rate variances

- Higher rates being paid than planned due to wage award
- Higher or lower grade of workers being used than planned.
- Payment of unplanned overtime or bonus

e. **Direct Labour Efficiency Variance**

This is the difference between the standard hours for the actual production

achieved and the hours actually worked expressed at the standard rate.

Possible causes of labour efficiency variances

- Use of incorrect grade of labour
- Poor workshop organization and supervision
- Incorrect materials and/or machines.

f. Idle Time Variance is determined as idle time x standard direct labour rate per hour.

Possible Causes of idle time variance

- Machine breakdown
- Power outages
- Work stoppages
- Ineffective co-ordination among staff

Variable Overhead Cost Variance

This is the difference between the actual overheads incurred and the variable overhead absorbed. This variance can further be analyzed overhead expenditure variance and variable overhead efficiency variance.

a. Variable Overhead Expenditure Variance:

This is the difference between the actual variable overheads incurred and the allowed variable overheads based on the actual hours worked.

b. Variable Overheads Efficiency Variance

This is the difference between the standard variable overheads and the allowed variable overhead for the actual production achieved. It is determined as: (Standard hours produced less Actual hours worked) x VOR per hour.